

Handout 02.11: Interest Math Practice

Name: _____ Date: _____ Partner: _____

Rates on this sheet are typical for 2026 and marked [update]. The people are made up.

My do now guess: a \$500 phone at \$25 a month on a store card will cost \$_____ in total.

Part A: Debit vs credit, side by side

	Debit card	Credit card
Whose money is it?		
When does it leave?		
Is there a bill?		
Is there interest?		
What if the card is stolen?		
Does it build a credit score?		

Credit is _____

Debt is _____

Interest is _____

Part B: Simple interest on \$500

The formula: $I = P \times r \times t$ (interest = amount borrowed x rate as a decimal x years)

Rates as decimals: 5 percent = 0.05. 15 percent = 0.15. 24 percent = 0.24.

Worked example: \$500 at 5 percent for 1 year. $\$500 \times 0.05 \times 1 = \25 . Total owed: $\$500 + \$25 = \$525$.

Rate [update]	What it is like	One year: interest	One year: total owed	Three years: interest	Three years: total owed
5 percent	a good car loan	$\$500 \times 0.05$ $\times 1 = \$$ _____	$\$$ _____	$\$500 \times 0.05$ $\times 3 = \$$ _____	$\$$ _____
15 percent	a credit union card	$\$500 \times 0.15$ $\times 1 = \$$ _____	$\$$ _____	$\$500 \times 0.15$ $\times 3 = \$$ _____	$\$$ _____
24 percent	a typical bank credit card	$\$500 \times 0.24$ $\times 1 = \$$ _____	$\$$ _____	$\$500 \times 0.24$ $\times 3 = \$$ _____	$\$$ _____

At which rate does three years of interest cost more than half the phone? _____

(Grade 6: one-year columns only.)

Part C: Two more loans

1. Ines borrows \$1,200 on a store card at 15 percent *[update]* and pays nothing for two years. Interest: \$ _____ x _____ x _____ = \$ _____. Total owed: \$ _____.
2. A used car loan: \$8,000 at 5 percent *[update]* for four years. Interest: \$ _____ x _____ x _____ = \$ _____. Total paid: \$ _____.
3. Which loan cost more in dollars? _____ Which cost more as a percent of what was borrowed? _____ How can both be true?

Part D: The minimum payment trap

Leo puts a \$500 phone on a credit card at 24 percent APR [update]. That is 2 percent a month. Each month the card adds 2 percent of the balance, then his payment comes off.

The first three months at \$25 a month (follow along on the board):

Month	Balance at start	Add 2 percent	Pay \$25	Balance at end
1	\$500.00	\$10.00	\$25.00	\$485.00
2	\$485.00	\$9.70	\$25.00	\$469.70
3	\$469.70	\$9.39	\$25.00	\$454.09

Three months in, Leo has paid \$75 and still owes \$454.09. Copy the finished results from the slide:

Plan	Months to pay off	Total interest	Total paid for the \$500 phone
\$25 a month (the minimum)			
\$50 a month			
Paid in full the first month			

My recommendation (two sentences, with the numbers):



My do now guess was \$ _____. The real number was \$ _____. I was off by \$ _____.

Part E: The credit score, in one paragraph

A credit score is a number from _____ to _____ that shows _____.

It goes up when I _____ and _____.

It goes down when I _____ or _____.

It matters because _____, _____, and sometimes _____ look at it.

Grade 8 stretch

1. Do months 4 and 5 of the \$25 plan by hand and check them against the pattern.
2. What monthly payment would clear the \$500 in ten months? (Try \$55, then \$56.)
About how much interest is that?
3. Three cards: a bank card at 24 percent, a store card at 29 percent, a credit union card at 15 percent [update]. A person with a good score should pick _____ because _____.

Exit card (3-2-1)

3 differences between credit and debit: _____, _____, _____

2 things that move a credit score: _____, _____

1 number: the total cost of the \$500 phone at \$25 a month: \$ _____

Teacher key

Part A: Debit: your money; right now; no bill; no interest; a stolen card drains your account; no. Credit: the bank's money, borrowed; when you pay the bill; yes, every month; yes, if not paid in full; you can dispute the charge and the bank's money is at risk; yes. Credit is borrowing money with a promise to pay it back. Debt is money you owe. Interest is the price of borrowing (or, on savings, the price the bank pays you; Lesson 2.12).

Part B: 5 percent: \$25, \$525; \$75, \$575. 15 percent: \$75, \$575; \$225, \$725. 24 percent: \$120, \$620; \$360, \$860. Three years costs more than half the phone (\$250) at 24 percent (\$360); at 15 percent it is \$225, just under.

Part C: 1. $\$1,200 \times 0.15 \times 2 = \360 ; total \$1,560. 2. $\$8,000 \times 0.05 \times 4 = \$1,600$; total \$9,600. 3. The car loan cost more in dollars (\$1,600 vs \$360); the store card cost more as a percent (30 percent of what was borrowed over two years vs 20 percent over four years). Both are true because the car loan is bigger and longer; the rate tells you the price per dollar per year, the dollars tell you the bill.

Part D: \$25 a month: 26 months; \$144.94 interest; \$644.94 total. \$50 a month: 12 months; \$63.50 interest; \$563.50 total. Paid in full: 1 month; \$0; \$500. Recommendation, "got it" example: "Pay \$50 a month because it is done in 12 months and costs \$63.50 in interest instead of \$144.94. Paying it off in full is better still if he has the \$500." Month-by-month check for the stretch: month 4: $\$454.09 + \$9.08 = \$463.17$ minus \$25 = \$438.17; month 5: $\$438.17 + \$8.76 = \$446.93$ minus \$25 = \$421.93.

Part E: 300 to 850; how well you have paid people back; pay on time every month, use a small part of the limit; pay late, max out a card (also applying for too much at once); a landlord, a car lender, an employer.

Grade 8 stretch: 2. About \$56 a month clears it in ten months; total paid about \$557, so interest about \$57. 3. The credit union card at 15 percent; the lowest rate costs the least if a balance is ever carried.

Exit: any three from Part A; any two from Part E; \$644.94.